ITEM 1

On a certain day a car started from rest and accelerated to $50ms^{-1}$ in 10 seconds. The driver maintained that velocity for 20 seconds and suddenly decelerated to rest in 2seconds causing him to crush into the windscreen.

The average speed in town should not exceed the speed limit of $8ms^{-s}$.At a time it came to rest, it had come in contact with a piece of wood that was in the road according to the equation $v_2 - v_1 = -(u_2 - u_1)$since the car separated from the piece of wood.

TASK

As a physics learner,

a) Help the driver to ascertain whether the speed limit was exceeded.
b) Advice the driver about how to avoid crashing into the windscreen again.
c) Help the senior four students at your school to prove the equation above stating all the conditions considered?
d) Help the driver to state the laws that govern the equation above?

ITEM 2

You are working as a design engineer at a manufacturing company. Your team is developing a new high precision weighing scale that will be used in laboratories to measure small masses accurately to ensure precision in its measurements before proceeding with design; you need to verify the consistency of formulas used in the calculation?

Task

a) Identify the fundamental and derived physical quantities require in the calibration of the weighing scale.
b) Use dimensional analysis to check whether the following equations used in scale design are dimensionally consistent.
   i. Force acting on the mass, F= mg.
   ii. The oscillation period of the scale's pan T =$2\pi\sqrt{l/g}$

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c) Explain why dimensional analysis is crucial in verifying these equations before building the prototype.

d) Discuss any potential design challenges that could arise due to errors in dimensions and how they can be corrected.

ITEM 3

In democratic Republic of Congo, there is war between the government and m23 rebels, due to the power war conflicts many people are suffering and they do not have what to eat. The united nations have decided to distribute food and other relief items using aircrafts.

An aircraft flying horizontally at 300km/hr at an altitude of 1000meters is dropping relief supplies to war victims in their camps. The package must land as close to the target as possible

Task

a) Help the war victims to know the time the package takes to reach the ground.

b) Help the war victims to know how far horizontally the package will travel before impact

c) Help the war victims to know how parachutes work and how they help the solidiers delivering the relief to come to the ground from the aircraft.

d) Explain how air resistance and terminal velocity affect the delivery of food items?

e) Given that one of the bags of food items of mass 2 tonnes collided with lorry of mass 3 tonnes.The bag was moving at $20ms^{-1}$ while the lorry was moving at $10ms^{-1}$ and after collision the lorry moved at $15ms^{-1}$.Calculate the velocity of the bag of food items after collision?

END

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